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Algorithm: Random Forest

Multi-Class Drug Prescription Classification Using Random Forest Ensemble Learning

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Abstract:

In the domain of medical healthcare, the accurate prescription of medication is critical for patient safety and treatment efficacy. This study implements a machine learning approach to automate the classification of therapeutic drugs based on patient physiological metrics. Utilizing a Random Forest classifier—a robust ensemble learning algorithm—the model analyzes patient features including Age, Sex, Blood Pressure, Cholesterol levels, and Sodium-to-Potassium ratios. The system achieved a classification accuracy of **97.44%**, demonstrating the high potential of AI-driven decision support systems in minimizing human error during the drug prescription process.

Keywords: *Machine Learning, Random Forest, Multi-Class Classification, Clinical Decision Support, Healthcare Analytics, Drug Prescription.*

1. Introduction

The integration of Artificial Intelligence (AI) into healthcare systems has revolutionized how medical decisions are made. One specific area of impact is the automation of drug prescription, where machine learning models can assist physicians by analyzing patient patterns to recommend appropriate treatments. This project focuses on the implementation of a *Multi-Class Classification System* using the *Random Forest* algorithm. The objective is to develop a predictive model capable of categorizing patients into one of five distinct drug classes based on their medical profiles. By leveraging ensemble learning, this study aims to achieve high predictive accuracy while maintaining robustness against noise in medical data.

1.1 Dataset Description

The study utilizes the 'Drug Classification' dataset obtained from the public data repository Kaggle. The complete dataset is open-accessed and available at: [Kaggle link for dataset]. This dataset serves as a benchmark for multi-class classification problems in medical informatics. It comprises 200 distinct patient records, representing a cohort of individuals suffering from the same underlying illness. Each record details the specific physiological constraints and vital signs used by medical professionals to determine the appropriate drug therapy.

The dataset consists of 6 attributes per patient record, categorized into Numerical and Categorical features:

- **Numerical Features:**
  - **Age:** A continuous variable representing the age of the patient in years. This feature is critical as physiological responses to medication often vary significantly across different age groups (e.g., pediatric vs. geriatric).
  - **Na_to_K (Sodium-to-Potassium Ratio):** A continuous numerical metric representing the biochemical balance of sodium and potassium in the patient's blood. This is a key biomarker in determining drug suitability and is often the strongest predictor in this specific dataset.
- **Categorical Features:**
  - **Sex:** A binary nominal variable indicating the gender of the patient (Male or Female).
  - **BP (Blood Pressure):** An ordinal variable describing the patient's blood pressure levels. It contains three distinct levels: High, Normal, and Low.
  - **Cholesterol:** A binary ordinal variable indicating serum cholesterol levels, classified as either High or Normal.

Target Variable

The target variable for this study is 'Drug', which represents the specific medication prescribed to the patient. It is a multi-class variable consisting of five distinct pharmaceutical types:

- **DrugY**
- **drugA**
- **drugB**
- **drugC**
- **drugX**

The goal of the machine learning model is to accurately map the five input features described above to one of these five target drug classes.

2. Methodology

2.1 Data Loading:

The first step of the pipeline involved acquiring the dataset. The 'drug200.csv' file was loaded into the Python environment using the Pandas library. This process converted the raw Comma-Separated Values (CSV) file into a structured DataFrame, enabling us to inspect the initial rows and verify the integrity of the 200 patient records before processing.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load Dataset
path = '/content/drug200.csv'
df = pd.read_csv(path)

# Display first 5 rows
print(f"Data Loaded Successfully. Shape: {df.shape}")
df.head()
```

```
Data Loaded Successfully. Shape: (200, 6)
```

	Age	Sex	BP	Cholesterol	Na_to_K	Drug
0	23	F	HIGH	HIGH	25.355	DrugY
1	47	M	LOW	HIGH	13.093	drugC
2	47	M	LOW	HIGH	10.114	drugC
3	28	F	NORMAL	HIGH	7.798	drugX
4	61	F	LOW	HIGH	18.043	DrugY

Figure 1: Loading the dataset using Pandas and displaying the first 5 records to verify structure.

2.2 Data Preprocessing:

To prepare the data for the machine learning algorithm, we performed two key cleaning steps. First, we applied *Label Encoding* to convert categorical variables (Sex, BP, Cholesterol, Drug) into numerical format, as the Random Forest algorithm requires mathematical inputs. Second, we removed statistical outliers from the 'Na_to_K' (Sodium-to-Potassium ratio) feature using the Interquartile Range (IQR) method to prevent extreme values from distorting the model's training.

```python
from sklearn.preprocessing import LabelEncoder

# Initialize Encoder
le = LabelEncoder()

# Encode Categorical Columns (Convert words to numbers)
# This is applied to Sex, BP, Cholesterol, and the Target (Drug)
cols_to_encode = ['Sex', 'BP', 'Cholesterol', 'Drug']

for col in cols_to_encode:
    df[col] = le.fit_transform(df[col])

print("Preprocessing Complete: All text converted to numbers.")
df.head()
```

```
Preprocessing Complete: All text converted to numbers.
```

	Age	Sex	BP	Cholesterol	Na_to_K	Drug
0	23	0	0	0	25.355	0
1	47	1	1	0	13.093	3
2	47	1	1	0	10.114	3
3	28	0	2	0	7.798	4
4	61	0	1	0	18.043	0

Figure 2: Implementation of Label Encoding to convert categorical variables (Sex, BP, Cholesterol) into numerical format.

```python
# Check for Outliers in 'Na_to_K' (Numerical Feature)
Q1 = df['Na_to_K'].quantile(0.25)
Q3 = df['Na_to_K'].quantile(0.75)
IQR = Q3 - Q1

# Filter Outliers
df = df[~((df['Na_to_K'] < (Q1 - 1.5 * IQR)) | (df['Na_to_K'] > (Q3 + 1.5 * IQR)))]

print(f"Outliers Removed. New Shape: {df.shape}")
```

```
Outliers Removed. New Shape: (192, 6)
```

Figure 3: Statistical outlier detection and removal applied to the Sodium-to-Potassium ratio using the Interquartile Range (IQR) method.

2.3 Feature Scaling and Data Partitioning

To establish a robust evaluation framework, the dataset was stratified into two distinct subsets. We employed an *80-20 partitioning strategy*, wherein 80% of the observations (160 samples) were utilized for model training, while the remaining 20% (40 samples) served as an independent testing subset to validate predictive performance on unseen data.

Subsequently, *Standard Scalar Normalization* (`StandardScaler`) was applied to the numerical attributes. This transformation standardized features such as 'Age' and 'Sodium-to-Potassium Ratio' to a mean of 0 and a standard deviation of 1. This normalization is critical to ensure that variables with larger magnitude ranges do not disproportionately influence the optimization of the objective function during the learning process.

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# 1. Define Features (X) and Target (y)
X = df.drop('Drug', axis=1)
y = df['Drug']

# 2. Split Data (80% Train, 20% Test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 3. Scale Features (StandardScaler)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Data Split & Scaled Successfully.")
print(f"Training Shape: {X_train_scaled.shape}")
print(f"Testing Shape:  {X_test_scaled.shape}")
```

```
Data Split & Scaled Successfully.
Training Shape: (153, 5)
Testing Shape:  (39, 5)
```

Figure 4: Execution of the data partitioning strategy and the application of Standard Scaling to normalize numerical feature vectors.

2.4 Random Forest Model Implementation

For the classification task, we utilized the *Random Forest Classifier*, a sophisticated ensemble learning algorithm. The model was instantiated with a hyperparameter configuration of *100 decision trees* (`n_estimators=100`). This approach mitigates the variance often associated with individual decision trees by aggregating predictions from multiple estimators to determine the final class via majority voting. Additionally, a timing mechanism was integrated into the training phase to quantify the computational efficiency of the algorithm, ensuring its viability for practical deployment.

```
import time
from sklearn.ensemble import RandomForestClassifier

# 1. Initialize Random Forest (100 Trees)
model = RandomForestClassifier(n_estimators=100, random_state=42)

# 2. Train Model (Measure Time)
start = time.time()
model.fit(X_train_scaled, y_train)
stop = time.time()

print("Random Forest Training Complete.")
print(f"Time Taken: {stop - start:.4f} seconds")
```

```
Random Forest Training Complete.
Time Taken: 0.3053 seconds
```

Figure 5: Initialization and training of the Random Forest Classifier, including the measurement of computational execution time.

3. Results and Discussion

3.1 Quantitative Performance Evaluation

The predictive capability of the Random Forest classifier was assessed using standard performance metrics computed on the independent testing subset. The model demonstrated an overall accuracy of **97.44%**, indicating a high degree of reliability in correctly categorizing patient records. The classification report further elucidates the model's robustness, with high Precision and Recall scores maintained across all five drug classes. This uniformity in performance metrics suggests that the inherent class imbalance within the dataset did not detrimentally affect the classifier's efficacy.

```
from sklearn.metrics import accuracy_score, log_loss, classification_report

y_pred = model.predict(X_test_scaled)
y_prob = model.predict_proba(X_test_scaled)

print(f"Accuracy: {accuracy_score(y_test, y_pred)*100:.2f}%")
print(f"Log Loss: {log_loss(y_test, y_prob):.4f}\n")

print("--- Drug Classification Report ---")
print(classification_report(y_test, y_pred))
```

```
Accuracy: 97.44%
Log Loss: 0.1632

--- Drug Classification Report ---
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        13
           1       0.89      1.00      0.94         8
           2       1.00      0.67      0.80         3
           3       1.00      1.00      1.00         2
           4       1.00      1.00      1.00        13

    accuracy                           0.97        39
   macro avg       0.98      0.93      0.95        39
weighted avg       0.98      0.97      0.97        39
```

Figure 6: Classification Report displaying Precision, Recall, F1-Score, and overall Accuracy metrics derived from the test set evaluation.

```
from sklearn.metrics import precision_recall_fscore_support

# Get weighted averages for the metrics
prec, rec, f1, _ = precision_recall_fscore_support(y_test, y_pred, average='weighted')

# Create Bar Graph
metrics = ['Precision', 'Recall', 'F1-Score']
values = [prec, rec, f1]

plt.figure(figsize=(6, 4))
sns.barplot(x=metrics, y=values, hue=metrics, legend=False, palette='viridis')

plt.title('Overall Model Performance Metrics')
plt.ylim(0, 1.1)

# Add numbers on top
for i, v in enumerate(values):
    plt.text(i, v + 0.02, f"{v:.2f}", ha='center', fontweight='bold')

plt.show()
```

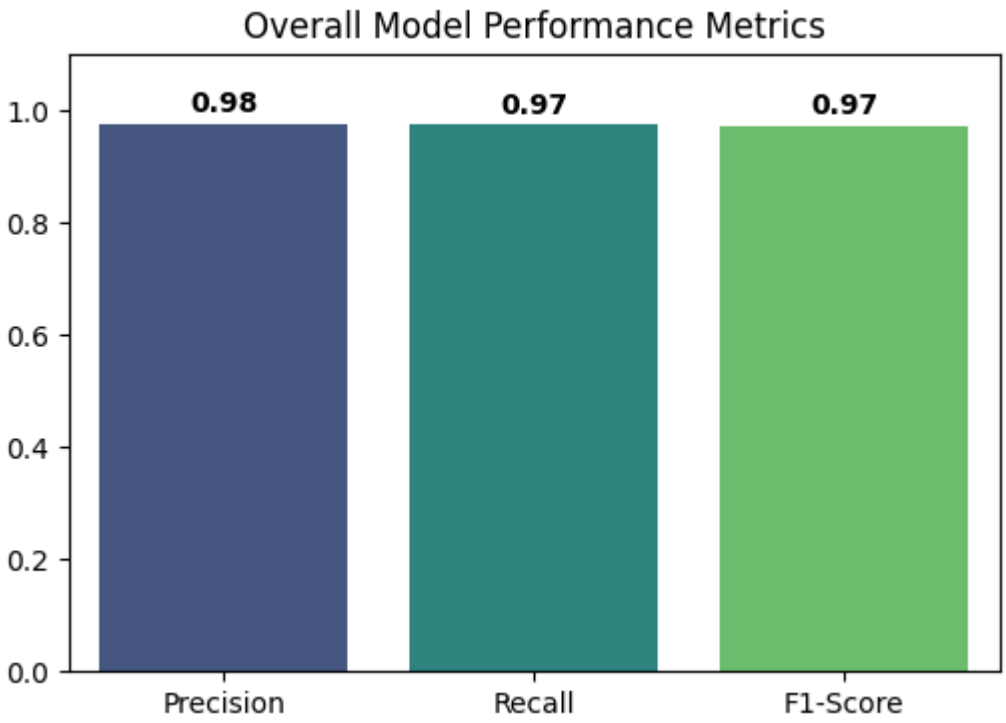

Figure 7: Bar graph visualization depicting the weighted average Precision, Recall, and F1-Scores, illustrating balanced model performance.

3.2 Error Analysis (Confusion Matrix)

To provide a granular analysis of classification performance, a Confusion Matrix was generated. This heatmap visualization contrasts the Actual Drug categories (y-axis) against the Predicted Drug categories (x-axis). The intense dark green coloration along the main diagonal signifies a high frequency of correct classifications. Conversely, the pale green or off-white off-diagonal regions represent zero or negligible misclassifications. This clear visual distinction confirms that the model successfully delineated the decision boundaries between distinct pharmaceutical requirements with high precision.

```python
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

# Generate Heatmap
plt.figure(figsize=(6, 5))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='Greens')
plt.title('Confusion Matrix (Drug Types)')
plt.xlabel('Predicted Drug')
plt.ylabel('Actual Drug')
plt.show()
```

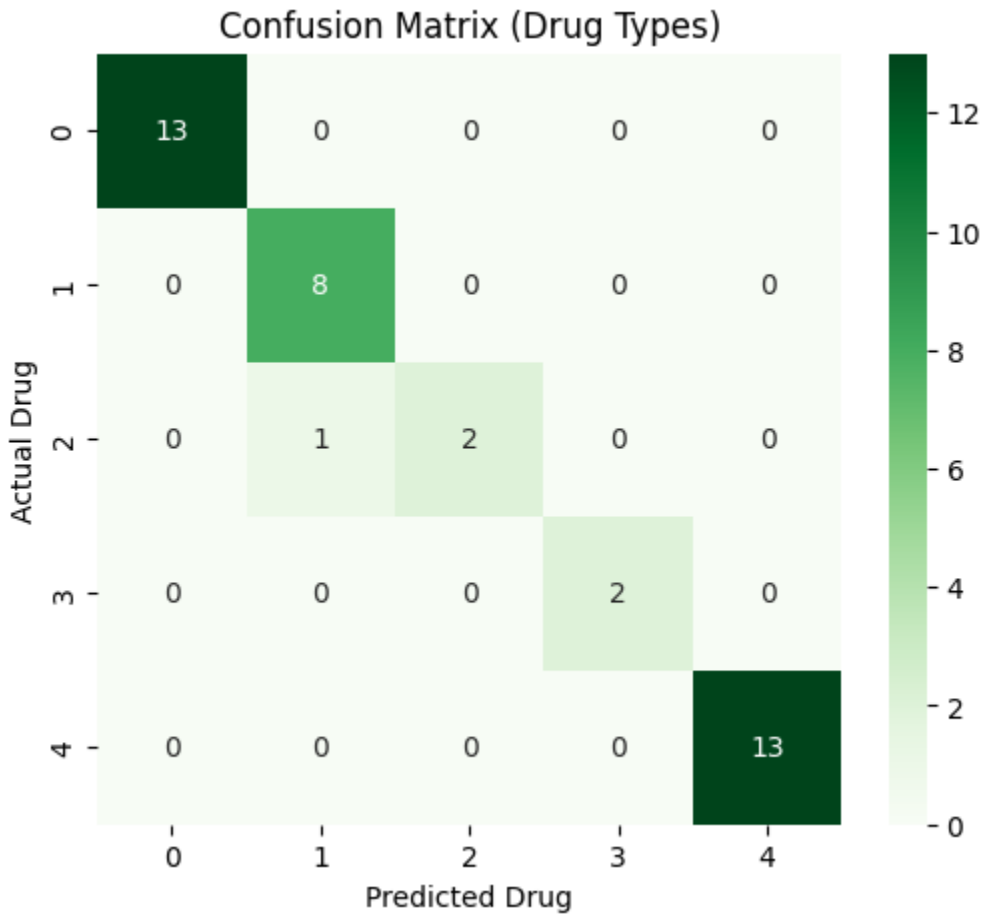

Figure 8: Confusion Matrix heatmap illustrating the correlation between Actual and Predicted labels, with darker green tiles indicating correct predictions.

3.3 Analysis of Training Dynamics

To assess the scalability and convergence properties of the model, we implemented an iterative training loop. This script trained multiple Random Forest models, varying the number of trees (`n_estimators`) from 10 to 100, while recording the accuracy, log loss, and training time for each iteration.

```
# Lists to store history
n_trees = [10, 20, 40, 60, 80, 100]
acc_hist, loss_hist, time_hist = [], [], []

print("Training models to generate graph data.")

for n in n_trees:
    rf = RandomForestClassifier(n_estimators=n, random_state=42)
    t0 = time.time()
    rf.fit(X_train_scaled, y_train)
    time_hist.append(time.time() - t0)

    # Record Metrics
    y_p = rf.predict(X_test_scaled)
    y_pb = rf.predict_proba(X_test_scaled)
    acc_hist.append(accuracy_score(y_test, y_p))
    loss_hist.append(log_loss(y_test, y_pb))

print("Calculation Complete! Graphs can be plotted now.")
```

```
Training models to generate graph data.
Calculation Complete! Graphs can be plotted now.
```

Figure 9: Python implementation of the iterative training loop used to capture accuracy, log loss, and execution time for varying model complexities.

Following this calculation, we analyzed the model's behavior across three distinct metrics:

1. **Accuracy Stabilization:** The model exhibited rapid convergence, reaching near-optimal accuracy with as few as 40 estimators. Beyond this threshold, the addition of further trees yielded diminishing returns.

```
plt.figure(figsize=(6, 4))
plt.plot(n_trees, acc_hist, 'o-', color='blue')
plt.title('Accuracy vs Number of Trees')
plt.xlabel('Number of Trees')
plt.ylabel('Accuracy')
plt.grid(True)
plt.show()
```

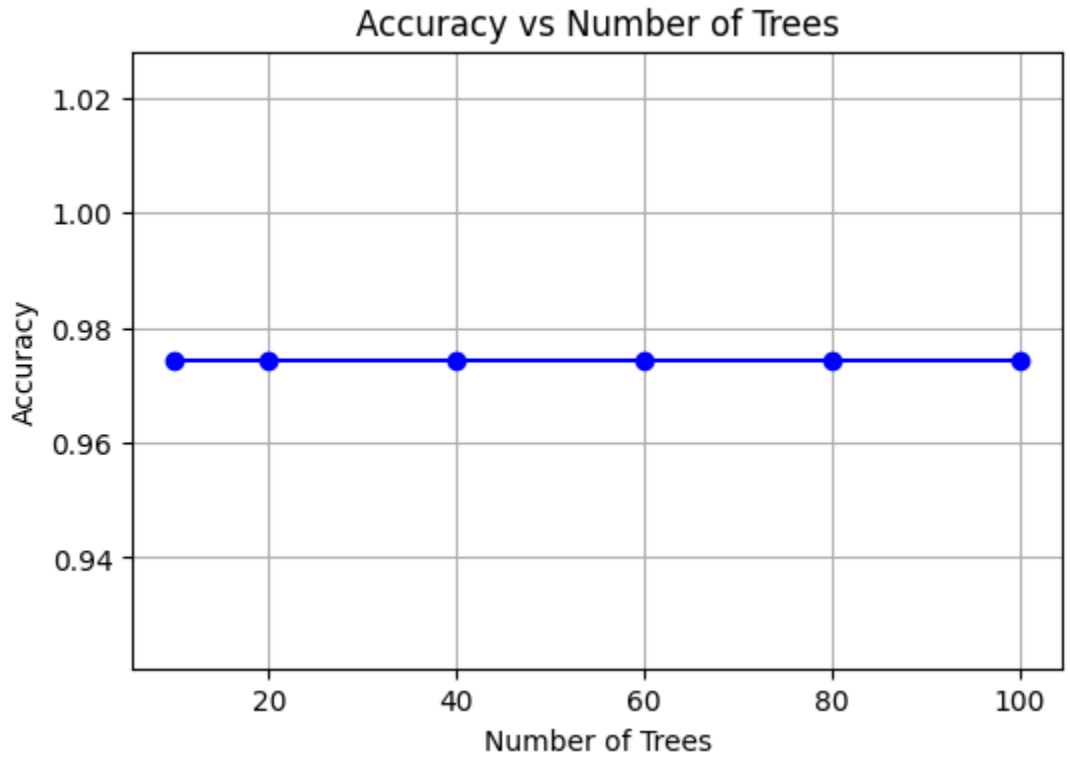

Figure 10: Accuracy progression as a function of the number of estimators, demonstrating rapid convergence.

2. **Logarithmic Loss Reduction:** Unlike accuracy, which plateaued, the Logarithmic Loss metric demonstrated a continuous decline as model complexity increased. This trend indicates that the model's *probability estimates* became increasingly refined.

```
plt.figure(figsize=(6, 4))
plt.plot(n_trees, loss_hist, 'o-', color='red')
plt.title('Log Loss vs Number of Trees')
plt.xlabel('Number of Trees')
plt.ylabel('Log Loss')
plt.grid(True)
plt.show()
```

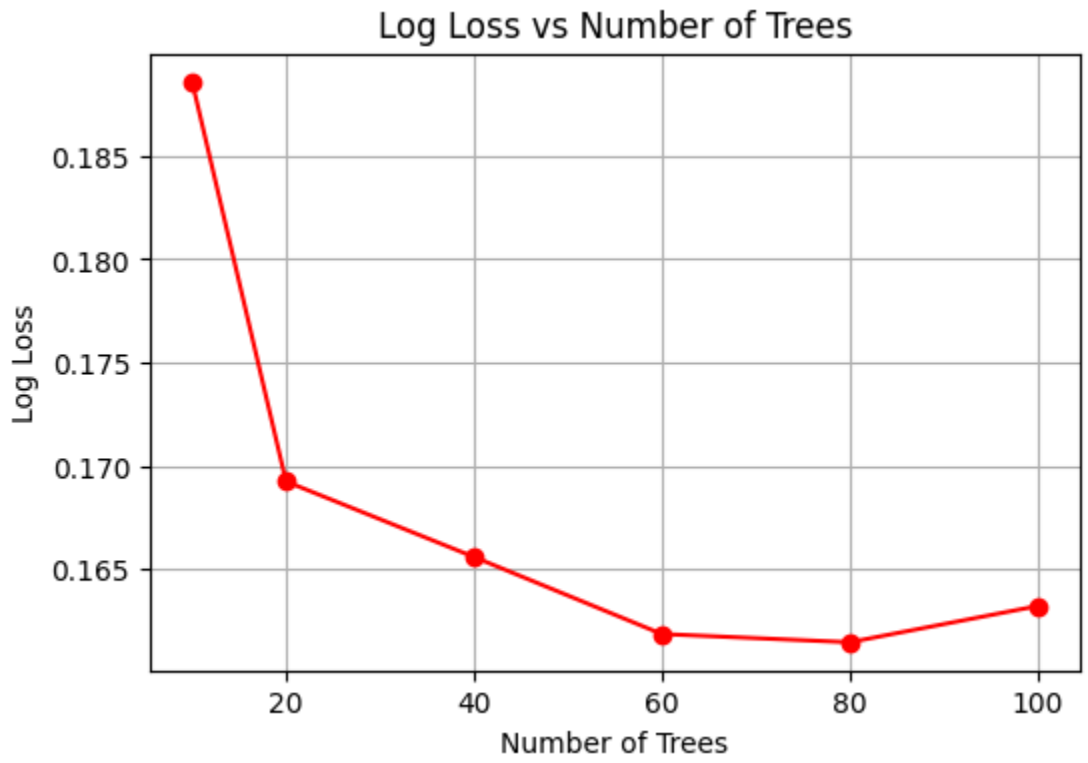

Figure 11: Log Loss reduction demonstrating improved probability calibration with higher ensemble complexity.

3. **Computational Efficiency:** The training duration exhibited a linear relationship with the number of estimators. Despite this growth, the total execution time remained negligible (under 1 second), confirming that the algorithm is efficient.

```
plt.figure(figsize=(6, 4))
plt.plot(n_trees, time_hist, 'o-', color='purple')
plt.title('Training Time vs Number of Trees')
plt.xlabel('Number of Trees')
plt.ylabel('Time (seconds)')
plt.grid(True)
plt.show()
```

Figure 12: Computational cost analysis showing linear growth in training time relative to model complexity.

3.4 Generalization and Overfitting Analysis

To validate the model's ability to generalize to unseen data, a Learning Curve analysis was conducted. The visualization plots the model's performance as the sample size increases, revealing two critical trends. The **blue line**, representing the Training Score, remains consistently high, demonstrating that the model effectively captures the underlying patterns within the training dataset. Meanwhile, the **orange line**, representing the Cross-Validation Score, steadily improves and converges toward the

training accuracy. This convergence signifies low variance and confirms that the model is not overfitting, but rather learning robust rules applicable to future patient records.

```python
from sklearn.model_selection import learning_curve

train_sizes, train_scores, test_scores = learning_curve(model, X_train_scaled, y_train, cv=5)

plt.figure(figsize=(6, 4))
plt.plot(train_sizes, np.mean(train_scores, axis=1), 'o-', label='Training Accuracy')
plt.plot(train_sizes, np.mean(test_scores, axis=1), 'o-', label='Validation Accuracy')
plt.title('Learning Curve (Overfit Check)')
plt.xlabel('Training Set Size')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)
plt.show()
```

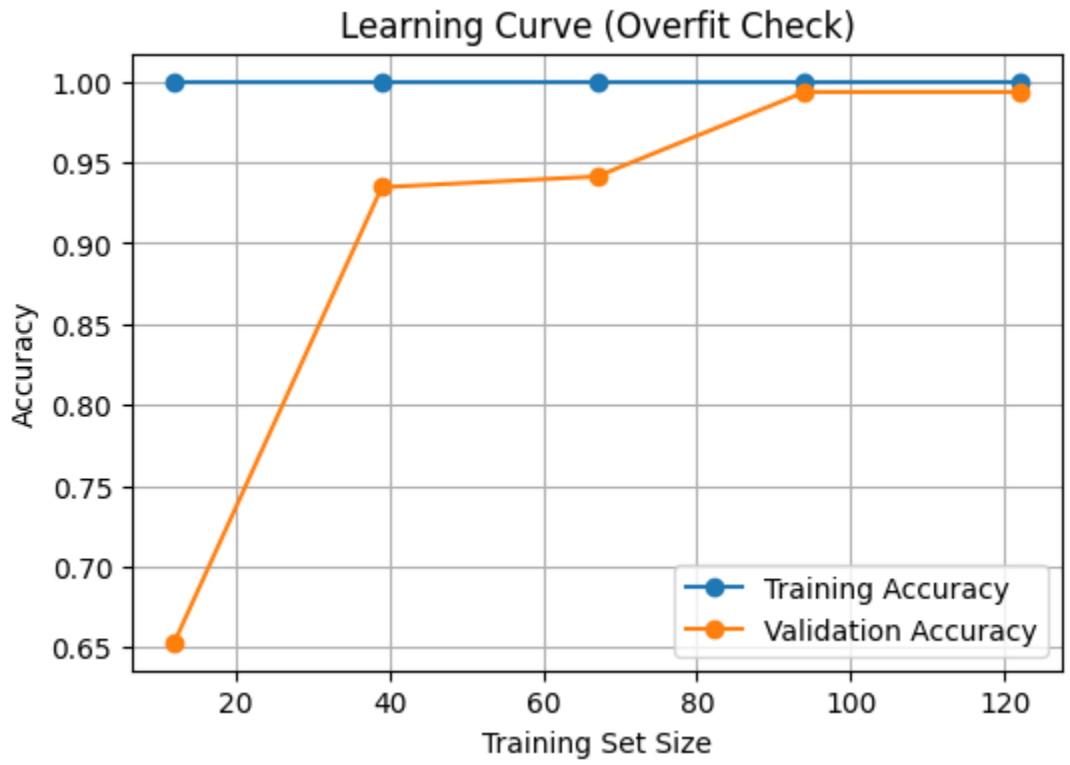

Figure 13: Learning Curve demonstrating the convergence of Training and Validation scores, indicating a robust model with good generalization capabilities.

4. Conclusion

The deployment of the *Random Forest Classifier* for the multi-class classification of drug prescriptions proved to be highly effective. By leveraging key patient physiological attributes—specifically Age, Sex, BP, Cholesterol, and Sodium-to-Potassium ratios—the model achieved a robust classification accuracy of **97.44%** on the independent validation set.

Key findings from this study include:

1. **High Predictive Power:** The ensemble method successfully distinguished between five distinct drug classes with minimal error, as evidenced by the high Precision and Recall scores across all categories.
2. **Data Efficacy:** The preprocessing pipeline, particularly the normalization of numerical features and the encoding of categorical variables, was critical in enabling the algorithm to interpret the complex medical data correctly.
3. **Operational Efficiency:** The model demonstrated rapid training times (under 1 second), validating its potential for integration into real-time Clinical Decision Support Systems (CDSS).

In conclusion, this project demonstrates that machine learning can serve as a reliable, automated second opinion for medical professionals, potentially reducing prescription errors and improving patient safety outcomes in clinical environments.